

PracticeFinalExam_VA (7)

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Math 340: Elementary Matrix and Linear Algebra

PRACTICE FINAL

Name: Key Section#: _____ NetID: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 120-minute exam. It consists of seven problems on eleven pages; the exam is six sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculator and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- **Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.**

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. Let C be a 3×3 matrix, and let

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 3 \\ 8 \\ 2 \end{bmatrix}$$

be the first, second, and third column of C , respectively.

(a) Compute the determinant of C .

$$\det(C) = \det \begin{bmatrix} 2 & 0 & 3 \\ 1 & 4 & 8 \\ 0 & 1 & 2 \end{bmatrix} = 2(8 - 8) + 0 + 3(1 - 0) = 3$$

(b) Is C invertible? If yes, what is the determinant of C^{-1} ? If no, explain why.

$$\text{Yes! } \det(C^{-1}) = \frac{1}{\det(C)} = \boxed{\frac{1}{3}}$$

(c) Let P be the parallelepiped spanned by $\mathbf{u}_1, -2\mathbf{u}_2, 6\mathbf{u}_3$. What is the volume of P ?

$$\det \begin{bmatrix} \vec{u}_1 & -2\vec{u}_2 & 6\vec{u}_3 \end{bmatrix} = -12 \det C = \boxed{-36}$$

$$\text{So the volume is } |-36| = 36$$

(d) Is $\mathbf{v} = \begin{bmatrix} 4 \\ 14 \\ 3 \end{bmatrix}$ in $\text{span}\{\mathbf{u}_1, 3\mathbf{u}_2\}$?

$$c_1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 12 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \\ 3 \end{bmatrix}$$

$c_1 = 2, c_2 = 1$ or

$$\begin{bmatrix} 2 & 0 & 4 \\ 1 & 12 & 14 \\ 0 & 3 & 3 \end{bmatrix} \xrightarrow{\text{redn}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\begin{cases} c_1 = 2 \\ c_2 = 1 \end{cases}$

(e) Do the vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ form a basis for $\mathbb{R}^3$?

Yes! they are linearly independent (since $\det(C) \neq 0$)
and there are 3 of them, so this makes them a basis.
"dim($\mathbb{R}^3$)"

2. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ be a matrix in $\mathbb{R}^{2 \times 2}$.

(a) Find the set W_1 of all $B \in \mathbb{R}^{2 \times 2}$ such that $AB = BA$.

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+2c & b+2d \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 2a+b \\ c & 2c+d \end{bmatrix}$$

$$a = a + 2c \Rightarrow c = 0$$

$$b + 2d = 2a + b$$

$$2c + d = d \Rightarrow c = 0.$$

$$2a = 2d \Rightarrow a = d.$$

so $W_1 = \left\{ \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} : a, b \in \mathbb{R} \right\}$

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(b) Let W_2 is a subspace in $\mathbb{R}^{2 \times 2}$ defined by

$$W_2 = \left\{ \begin{bmatrix} a & 0 \\ b & a-b \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

Claim: Find a basis for W_2 .

A_1 A_2
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}$ this is a basis, b/c they span W_2 :

$\begin{bmatrix} a & 0 \\ b & a-b \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}$, and they're linearly independent, since A_1 is not a mult. of A_2 .

(c) What is the dimension of W_2 ?

$\boxed{2}$ b/c my basis has 2 elements.

$$\dim(P_2) = 3$$

3. Let $L: P_2 \rightarrow P_2$ be a linear transformation defined by

$$L(at^2 + bt + c) = (b-c)t^2 + (c-a)t + (a-b).$$

(a) Find $\dim \text{range } L$.

see (b): $\dim \ker L = 1$, so
 $\dim \text{range } L = 3 - 1 = \boxed{2}$

(b) Find a basis for $\ker L$.

$$L(at^2 + bt + c) = 0 = \underbrace{(b-c)}_{=0}t^2 + \underbrace{(c-a)}_{=0}t + \underbrace{(a-b)}_{=0}$$

$$\begin{array}{c} a \quad b \quad c \\ \left[\begin{array}{ccc|c} 0 & 1 & -1 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{array} \right] \xrightarrow{r_1 \leftrightarrow r_3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right] \xrightarrow{r_1 + r_2, r_2} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

$\downarrow r_2 + r_3 \rightarrow 0$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\downarrow (-1)r_2 + r_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\downarrow -r_2 \rightarrow r_2$

$$\left[\begin{array}{ccc|c} a & b & c \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Kernel is made of elements of the form

$$rt^2 + rt + r1 = r(t^2 + t + 1)$$

Basis: $t^2 + t + 1$

$$\begin{array}{l} c=r \\ a-r=0 \rightarrow a=r \\ b-r=0 \rightarrow b=r \end{array}$$

4. Let E, G, H all be 3×4 matrices.

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \text{ rank at most 3.}$$

(a) Is it possible that E has column rank 4? Justify your answer.

No, rank = column rank, and there are at most 3 pivots/leading ones, so column rank ≤ 3 .

(b) Is it possible that G has row rank 1? Justify your answer.

sure! let

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad 1 \text{ pivot} \Rightarrow \text{row rank} = 1.$$

(c) Suppose that you compute the row echelon form of matrix H and it has the form

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Find the solution(s) to $H\mathbf{x} = \mathbf{0}$, if any solutions exist. (Use x_1, x_2, x_3, x_4 for the unknowns of the linear system.)

$$x_4 = 0$$

$$x_3 + 4x_4 = 0$$

$$x_3 = -4x_4 = 0.$$

$$x_2 = r$$

$$x_1 + x_2 - 2x_3 - 3x_4 = 0$$

$$x_1 = -r$$

$$r \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ is my soln. set}$$

(d) Let $f: \mathbb{R}^s \rightarrow \mathbb{R}^t$ be a matrix transformation defined by $f(\mathbf{u}) = E\mathbf{u}$.

i. What are s and t ?

ii. Is f a linear transformation?

$$(i) \quad \begin{matrix} E \text{ is } 3 \times 4 \\ 3 \times 4 \end{matrix} \quad E \vec{u} \rightarrow \vec{u} \text{ is } 4 \times 1, \quad E \vec{u} \text{ is } 3 \times 1$$

$$\downarrow \qquad \qquad \downarrow$$

$$s=4 \qquad \qquad t=3$$

(ii) Yes! b/c f is a matrix transformation

5. Is the matrix

$$A = \begin{bmatrix} 1 & -5 & 0 \\ 1 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

diagonalizable? Justify your answer.

Let's find the eigenvalues of A :

$$\begin{aligned} \det(\lambda I_3 - A) &= \det \begin{bmatrix} \lambda-1 & 5 & 0 \\ -1 & \lambda-7 & 0 \\ 0 & 0 & \lambda-1 \end{bmatrix} = (\lambda-1)((\lambda-1)(\lambda-7)+5) \\ &= (\lambda-1)(\lambda^2 - 8\lambda + 7 + 5) \\ &= (\lambda-1)(\lambda^2 - 8\lambda + 12) \\ &= (\lambda-1)(\lambda-2)(\lambda-6) \\ &\quad \lambda = 1, 2, 6. \end{aligned}$$

Yes!! It's diagonalizable b/c it's a 3×3 matrix w/ 3 eigenvalues.

6. Let $L: P_1 \rightarrow \mathbb{R}^2$ be a linear transformation defined by

$$L(at + b) = \begin{bmatrix} 3b - a \\ 2a + b \end{bmatrix}.$$

Let $S_1 = (1, t)$ be the standard basis of P_1 , and T_1 be the standard basis of $\mathbb{R}^2$. Furthermore, let $S_2 = (3t + 5, t + 2)$ be another basis of P_1 and $T_2 = \left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right)$ be another basis for $\mathbb{R}^2$.

(a) Find $P_{S_1 \leftarrow S_2}$ $\rightarrow$ This means: Put S_2 vectors into S_1 coordinates.

$$[\vec{u}_1]_{S_1} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \quad [\vec{u}_2]_{S_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{so } P_{S_1 \leftarrow S_2} = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$$

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(b) Find $P_{S_2 \leftarrow S_1} = P^{-1} = \frac{1}{5-6} \begin{bmatrix} 1 & -2 \\ -3 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}$

$\rightarrow$ or

$$L(\vec{u}_1) = \begin{bmatrix} 15 - 3 \\ 6 + 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 11 \end{bmatrix}^{\vec{w}_1}$$

$$\begin{bmatrix} -1 & 2 & 12 \\ -1 & 1 & 11 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & 1 \end{bmatrix} \rightsquigarrow [\vec{w}_1]_{T_2} = \begin{bmatrix} -10 \\ 1 \end{bmatrix}$$

$$L(\vec{u}_2) = \begin{bmatrix} 6 - 1 \\ 2 + 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}^{\vec{w}_2}$$

$$\begin{bmatrix} -1 & 2 & 5 \\ -1 & 1 & 4 \end{bmatrix} \xrightarrow{8} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \end{bmatrix} \rightsquigarrow [\vec{w}_2]_{T_2} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$B = \begin{bmatrix} -10 & -3 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 12 \\ -1 & 1 & 11 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & 1 \end{bmatrix} \rightsquigarrow [\vec{w}_1]_{T_2} = \begin{bmatrix} -10 \\ 1 \end{bmatrix}$$

$$L(\vec{u}_2) = \begin{bmatrix} 6-1 \\ 2+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix} \vec{w}_2$$

$$B = \begin{bmatrix} -10 & -3 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 5 \\ -1 & 1 & 4 \end{bmatrix} \xrightarrow{+} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \end{bmatrix} \rightsquigarrow [\vec{w}_2]_{T_2} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

(c) What is the representation of L with respect to S_2 and T_2 ?

rep. of L wrt. S_1 and T_1 is: $L(1) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ $L(t) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$$Q_{T_1 \leftarrow T_2} = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} \rightarrow Q^{-1} = \frac{1}{-1+2} \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \quad A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{array}{ccc} S_1 & \xrightarrow{A} & T_1 \\ \uparrow P & & \downarrow Q^{-1} \\ S_2 & \xrightarrow{B} & T_2 \end{array}$$

$$B = Q^{-1}AP = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix} \\ = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 12 & 5 \\ 11 & 4 \end{bmatrix} = \begin{bmatrix} -10 & -3 \\ 1 & 1 \end{bmatrix}$$

7. (3+3+4+5 points)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 7 \\ -2 \\ -1 \end{bmatrix}$$

be two vectors in $\mathbb{R}^3$.

(a) Find the distance between the points whose position vectors are $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$\|\vec{v}_1 - \vec{v}_2\| = \left\| \begin{bmatrix} -6 \\ 5 \\ -1 \end{bmatrix} \right\| = \sqrt{6^2 + 5^2 + 1^2} = \sqrt{36 + 25 + 1} \\ = \sqrt{62}$$

(b) Let θ be the angle between $\mathbf{v}_1$ and $\mathbf{v}_2$.

Find $\cos \theta$.

$$\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\| \|\vec{v}_2\|} = \frac{(1)(7) + (3)(-2) + (-2)(-1)}{\sqrt{14} \sqrt{54}}$$

$$\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{\|\vec{v}_1\| \|\vec{v}_2\|} = \frac{(1)(7) + (3)(-2) + (-2)(-1)}{\sqrt{1^2 + 3^2 + 2^2} \sqrt{7^2 + 2^2 + 1^2}}$$

$$= \boxed{\frac{3}{\sqrt{14} \sqrt{54}}}$$

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(c) Let $\mathbf{w}_1 = \begin{bmatrix} 1 \\ 3 \\ a \end{bmatrix}$ and $\mathbf{w}_2 = \begin{bmatrix} 7 \\ a \\ -1 \end{bmatrix}$. Find a if $\mathbf{w}_1 \perp \mathbf{w}_2$.

$$\vec{w}_1 \cdot \vec{w}_2 = (1)(7) + (3)(a) + (a)(-1)$$

$$= 7 + 3a - a = 0$$

$$7 + 2a = 0 \rightarrow \boxed{a = -\frac{7}{2}}$$

(d) Let $\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ be vectors from $\mathbb{R}^3$, and define $\mathbf{a} \odot \mathbf{b}$ by

$$\mathbf{a} \odot \mathbf{b} = a_1 b_1 - a_2 b_2 + a_3 b_3.$$

Is the operation $\odot$ an inner product in $\mathbb{R}^3$?

$$\vec{a} \odot \vec{a} = a_1 a_1 - a_2 a_2 + a_3 a_3 = a_1^2 - a_2^2 + a_3^2 \stackrel{?}{\geq} 0 \text{ Not necessarily!}$$

$$\text{Pick } \vec{a} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ then } \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \odot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0 - 1^2 + 0 = -1 < 0 \therefore$$

Once ① failed,
we could stop.

So this violates condition (a) $\Rightarrow$ not an inner product.

Note: (b) $\vec{a} \odot \vec{b} = a_1 b_1 - a_2 b_2 + a_3 b_3 = b_1 a_1 - b_2 a_2 + b_3 a_3 = \vec{b} \odot \vec{a}$ is true.

$$\begin{aligned} \text{(c)} \quad (\vec{a} + \vec{b}) \odot \vec{c} &= \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix} \odot \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = (a_1 + b_1)c_1 - (a_2 + b_2)c_2 + (a_3 + b_3)c_3 \\ &= a_1 c_1 + b_1 c_1 - a_2 c_2 - b_2 c_2 + a_3 c_3 + b_3 c_3 \end{aligned}$$

(d)

$\mu \in \mathbb{R}$:

$$(\mu \vec{a}) \odot \vec{b} = (\mu a_1)b_1 - (\mu a_2)b_2 + (\mu a_3)b_3$$

$$= \mu(a_1 b_1 - a_2 b_2 + a_3 b_3)$$

$$= \mu(\vec{a} \odot \vec{b}) \text{ is true.}$$

$$\begin{aligned} &= a_1 c_1 - a_2 c_2 + a_3 c_3 \\ &\quad + b_1 c_1 - b_2 c_2 + b_3 c_3 \\ &= \vec{a} \odot \vec{c} + \vec{b} \odot \vec{c} \text{ is true.} \end{aligned}$$